

ADAPTIVE IMMUNE RESPONSE

Activation of B and T cells

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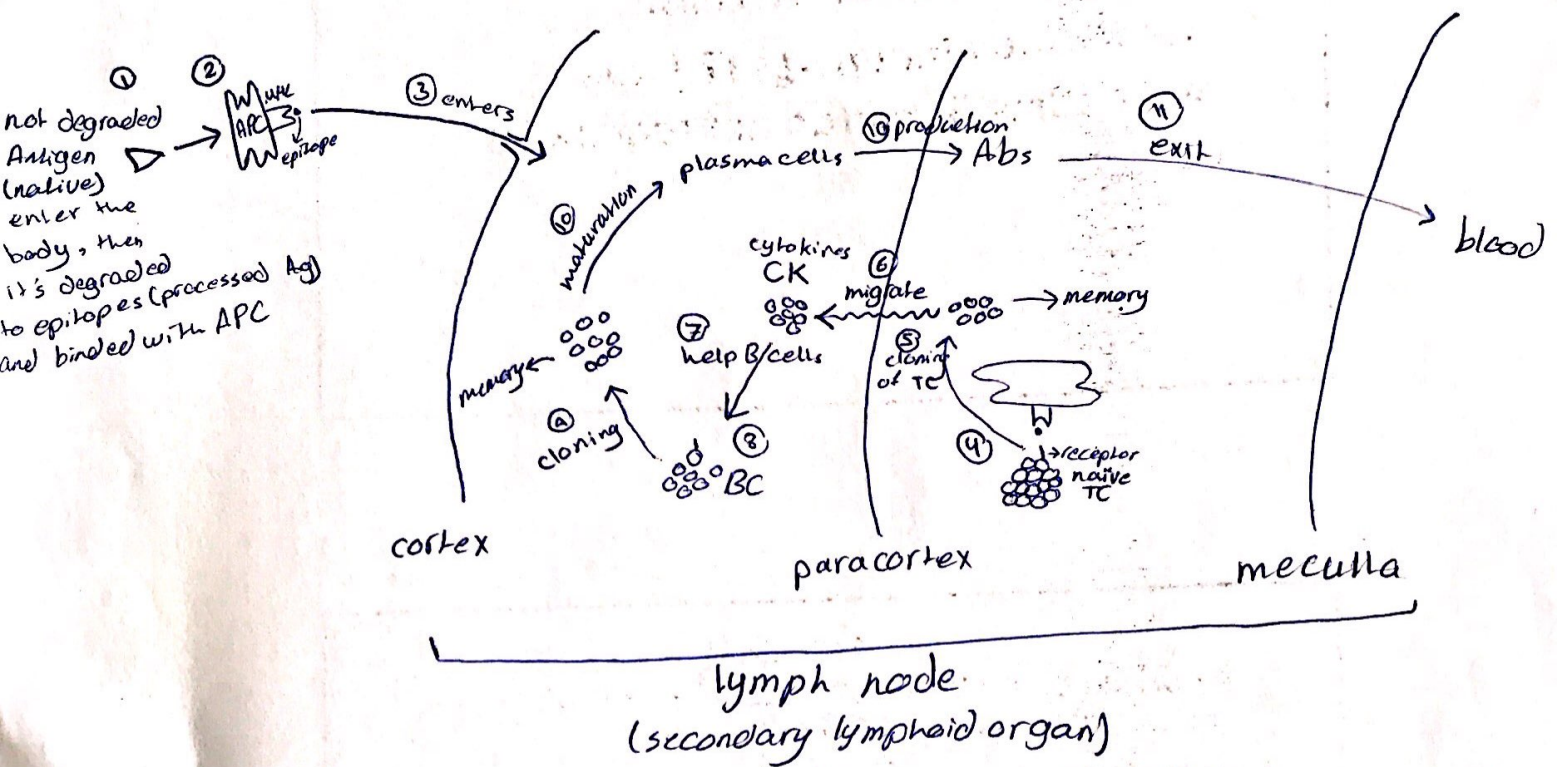
Lecture outline

Activation of B cells

- A. Dendritic cell maturation in response to a pathogen
- B. Activation of CD4+ cells
- C. Activation of B cells
- D. Cooperation between selected activated CD4+ and B cells
- E. Clonal expansion and differentiation of activated B cells to plasma cells
- F. Events in germinal centers
- G. T-independent B cell response
- H. Other ways to activate T cells
- F. Modulation of B cell activity

Activation of T cells

- A. Activation and Function of CD8+ T cells
- B. CD4-Independent activation of CD8+ T cells
- C. CD4-dependent activation of CD8+ T cells
- D. Killing of target cells by CD8+ T cells
- E. Termination of the CD8+ T cell response
- F. The traffic of activated CD8+ T cell
- G. $\gamma\delta$ cells and NKT cells



* B cells can't produce antibodies without the help of T cells (CD4TC)

④ only one T cell is selected to bind the epitope and MHC with TCR

⑤ the specific T cell is activated and cloned (gene and protein level), a part of the cells becomes memory cells

⑧ a B cell is selective against the epitope (without MHC), the selective B cells activated by the selective T cell using cytokines

T-dependent and T-independent Immune Response

- **Thymus-dep. (TD) Ags:** Most Ags are known as because the responding B cells require the help of Th cells.
- **The thymus-indep Ags (TI):** the responding B cells do not require Th cells.

T-dep Ags

Dendritic cell maturation in response to a pathogen

- **Professional or specialized APC**

Include: Dendritic cells, Macrophages and B cells

APC are found at all entry sites of Ags (airways, gastrointestinal tract or skin) and in lymphoid organs.

2 subsets of dendritic cells have been identified:

innate immunity ← plasmacytoid synthesize the IFN α and β in the early phase of i.r. (innate immunity).
present the Ag ← Myeloid dendritic cells have a major role in the induction of T cell i.r.

- **Characteristics of immature naïve dendritic cells:**

- a) present at all entry sites of the Ag and secondary lymphoid organs
- b) express pattern recognition receptors such as TLR (*follicle receptor*)
- c) Have large surface area
- d) express phagocytic receptors to enhance engulfment of bacteria

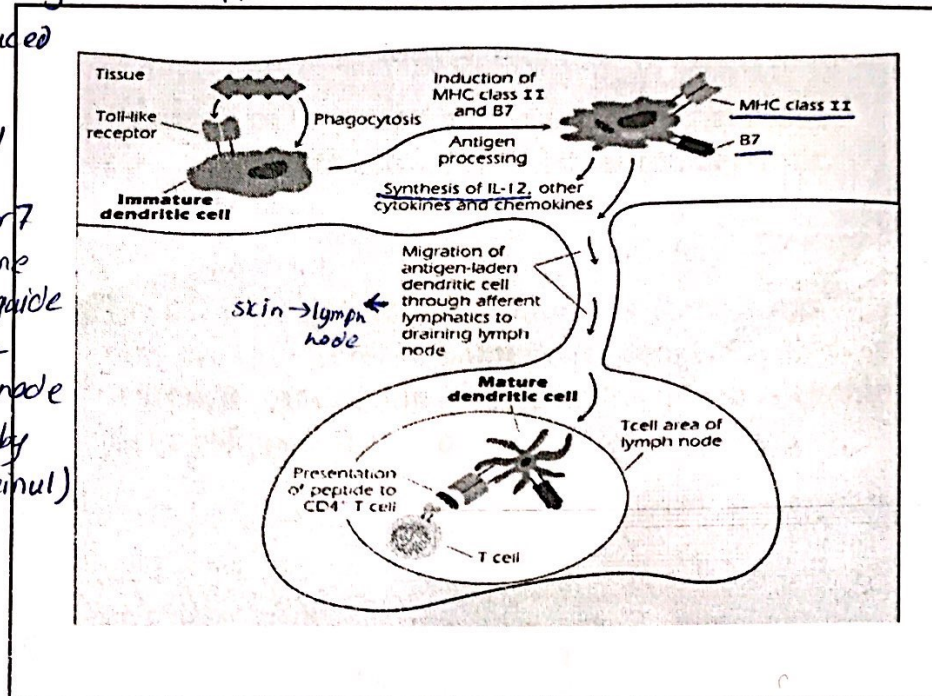
three major changes will happen!

1- MHC^{class II} will be produced by dendritic cells

naïve dendritic cell will be activated

2- chemokine receptor 7 (CCR7) and selectins and integrins will guide the antigen through migration to lymph node (CCR7 is expressed by high endothelial venule)

3- presentation



* harmless Ags do not activate B cells

• Maturation of naïve dendritic cells

1. attachment to the surface receptors
2. Engulfment of the Ag e.g bac
3. Ag processing: catabolizing the proteins into peptides in the endosomes following MHC class II pathway.
4. This induce
 - a. the expression of high levels of **MHC class II** and costimulatory molecules particularly **B7**
 - b. secretion of many cytokines including **IL-12**
 - c. upregulates the expression of chemokine receptor **CCR7**. CCR7L is expressed by high endothelial venules at the entrance of lymph nodes, the Ag-stimulated dendritic cells leave the tissue and migrate via lymphatics to the lymph node

- Dendritic cell with self molecules in normal tissue does not lead to activation of either the dendritic cell or T cells

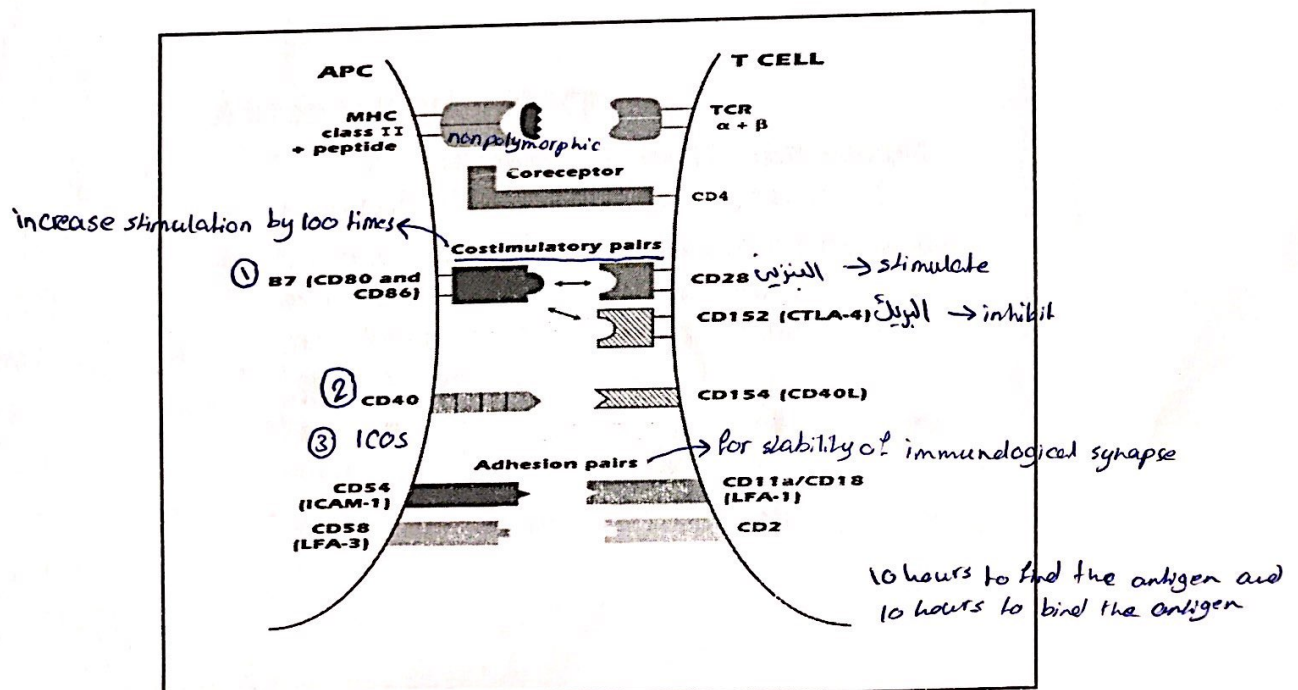
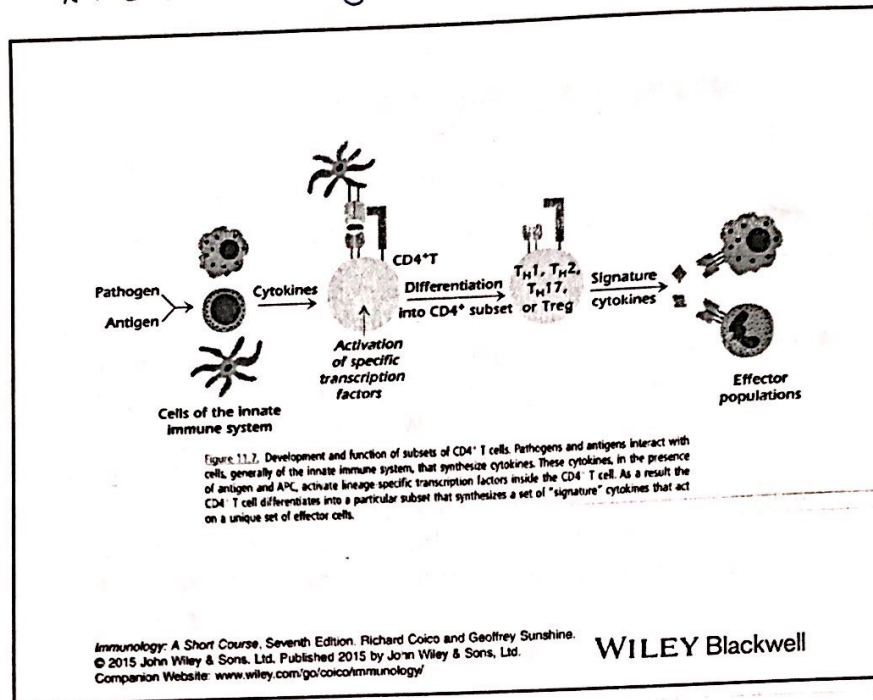
- T cells and B cells to many harmless foreign Ags may evoke little or no immune response. For this reason, in vaccination an adjuvant is mixed with the Ag to enhance activation of APC

Activation of CD4+ T cells

1- Interaction of APC and CD4+ T cells

- Activation of CD4+ cells occurs in the T cell area of L. node
- occurs within 8-10 hours of Ag exposure

- * the first signal of immunity is the selection of T cells against Ag and MHC
- * the second signal is co-receptors binding with MHC



in a procedure, a disturbance of the stability of immunological synapse happened $\rightarrow$ no immune response

Formation of immunological synapse

- The contact area between APC-peptide with CD4+ T cell is known as immunological synapse (i.s).
- i.s is required for sustained intracellular signaling and lasting about 8 hrs.
- Involve all costimulatory and adhesion molecules
- i.s is dynamic ^{حرکی}: its composition and structure change with time after the initial contact e.g ICAM-1 and LFA-1 are found in different regions at different times (this is the signal)
- In the T cell membrane, the structure of lipid is not homogenous and form lipid rafts enriched in cholesterol and glycosphingolipid. Once T cell is activated, Lipid rafts are mobilized to the synapse participating in the intracellular signaling

activation signal
for inside cell enzyme
machinery

A-Peptide/MHC and TCR

- The first signal involves interaction between the Ag-specific $V\alpha+V\beta$ of TCR with peptide/MHC.
- TCR detects very small number of foreign peptides as low as 4 peptide molecules out of thousands of peptides expressed on an APC, so the T cell is very sensitive to foreign Ag.
- The interaction is of low affinity so it is necessary but not sufficient for the activation
- The paired interaction enhance the affinity

4 peptides are enough to
activate the first signal

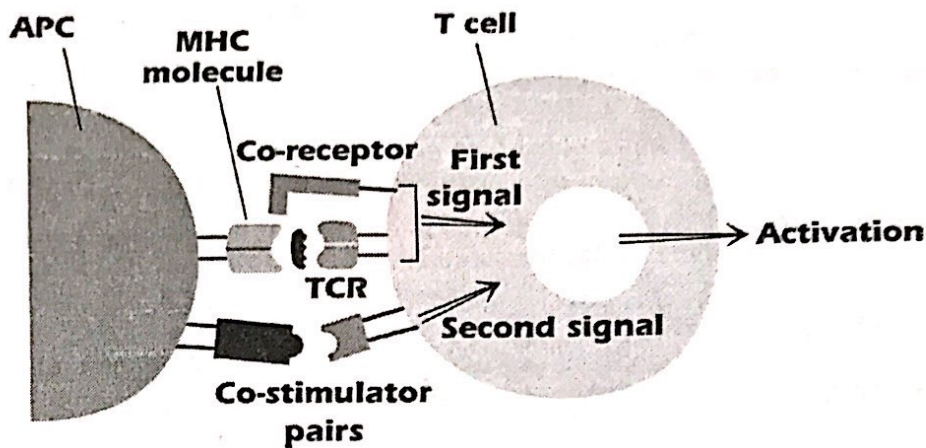


Figure 11.1. Activation of the naive T cell requires two signals. First signal = TCR with peptide + MHC; T cell co-receptor (CD4 or CD8) with MHC is a component of the first signal. Second signal = co-stimulator pairs at the APC-T-cell surface.

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B-MHC class II with CD4 coreceptor

The interaction of the nonpolymorphic part of an MHC class II molecule with the coreceptor CD4 greatly enhance the ability of a T cell to respond to Ag and hence lowers the T cell Ag stimulating threshold.

in the absence of coreceptor only a weak reaction happens

The interaction makes a cell 100 folds more responsive to Ag than in the absence of the interaction

Also CD4 plays an important role in signal transduction

c-Costimulatory Pairs

Deliver second signals that enhance and sustain the first signal
Ags that do not induce high levels of costimulator molecules do not activate T cells

B7 (APC) -CD28 (expressed constitutively on T cells)

In the absence of B7-CD28 interaction, the first signal alone not only is insufficient to activate the T cell but also results in T cell anergy

CD40 (APC)-CD40L (activated T cell):

The interaction upregulates T cell expression of CD40L and increase expression of B7 *CD40 do class switching*

ICOS ligand (APC)-ICOS (T cells) necessary for the formation of germinal centers

D-Adhesion molecules

Strengthen and stabilize the interaction

ICAM-1 (APC)...LFA-1 (T cells)

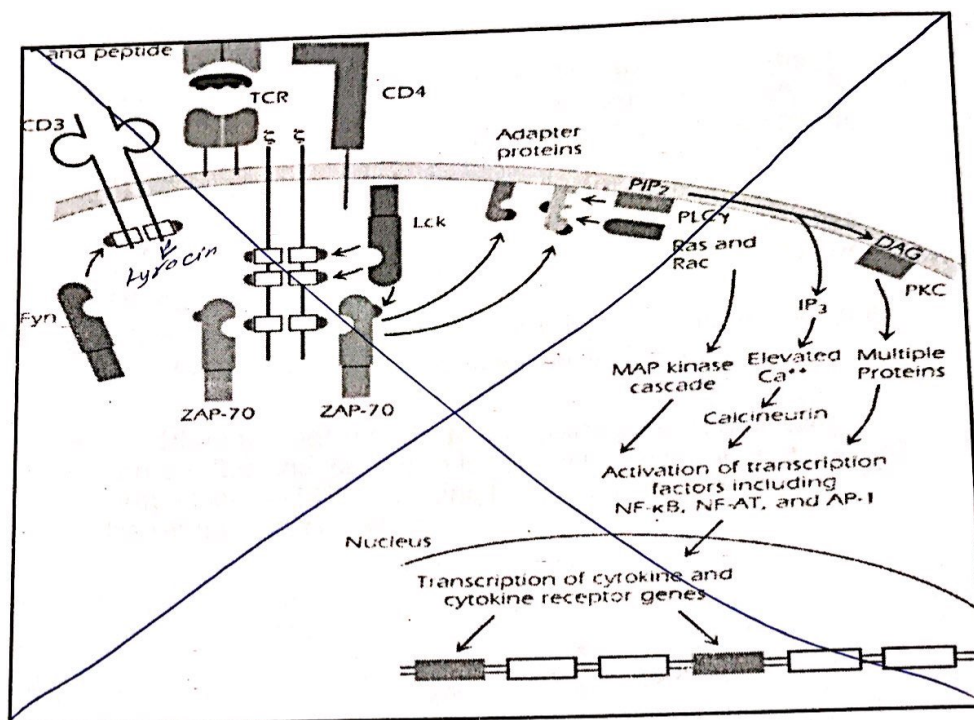
LFA-3....CD2 (T cells)

~~* ICOS molecule is necessary for formation of germinal centers~~

2. Intracellular events in CD4+ T cell activation

- The initial signal may involve the aggregation of multiple TCR molecules in the cell membrane.
- This triggers multiple intracellular cascades that spread from the membrane through the cytoplasm and into the nucleus.
- Some events occur within seconds, others within min. and others within hrs.
- The T cell changes its pattern of gene expression from resting stage to an activated state.

* Tyrosine is susceptible to phosphorylation (addition of phosphate)



- After binding of TCR, activation of CD45 mem protein Tyr phosphatase occurs which in turn activates within seconds of Tyr kinases, fyn (associated with the cytoplasmic tail of CD3) and lck (associated with the cytoplasmic tail of CD4). Both fyn and lck belong to src family of Tyr kinases.
- The phosphorylated ITAM in CD3 then act as docking sites for another Tyr Kinase ZAP-70 (syk family). Activation of ZAP-70 is a critical step.
- 2 adaptor molecules become phosphorylated thus a multiprotein complex of s. transduction molecules is assembled in sequence
- The adaptor molecules bind phospholipase c-γ which in turn catalyze the breakdown of the mem phospholipid phosphatidylinositol biphosphate (PIP2) into diacylglycerol (DAG) and inositol triphosphate (IP3).

- DAG activates the mem-associated Protein kinase C (PKC) which in turn activates a cascade of kinases, ultimately leading to the activation of cytoplasmic transcription factor, NF- κ B.
- IP3 increases the free Ca levels, which in turn activates the calcineurin protein which activates the transcription factor NF-AT.
Cyclosporin A and tacrolimus used to prevent graft rejection bind to calcineurin and prevent T cell activation.
- In addition, adaptor molecules activate guanosin binding proteins, Ras and Rac which in turn activate mitogen-activated protein kinase (MAP) leading to the activation of the transcription factor AP-1.

* drugs: cyclosporin A and tacrolimus are used to prevent graft rejection in transplantation as it prevent TC activation (without it transplantation fails)

3-Subsets of CD4+ T cells (T_H)

- 4 major subsets of CD4+ cells are known according to the secreted cytokines:
 T_{H0} , T_{H1} , T_{H2} , T_{H17} and T_{reg} cells (CD4 on the surface)
(depends on the antigen)
- T_{H0} cells: Have some features of both T_{H1} and T_{H2} cells, synthesizing IL-2, IL-4 and IFN- γ
- T_{H1} cells: $\rightarrow$ some kinds of bacteria
As a response to intracellular pathogens, dendritic and NK cells produce IL-12 which in turn activates the transcription factor Stat4 and Stat1 & T-bet
synthesize IFN- γ , IL-2, TNF- β (lymphotoxin)
these cytokines activate Macrophages, CD8+ T cells and NK cells to kill the intracellular pathogens Characteristics of CMI.
IFN- γ influences B cell class switching to Ig isotypes.

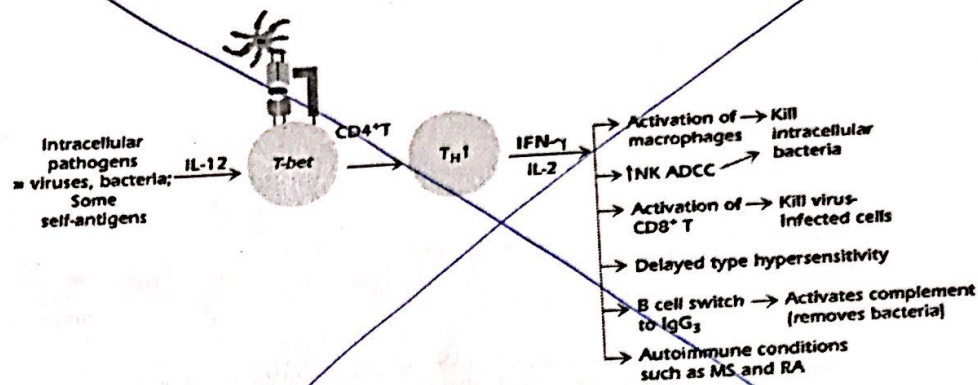


Figure 11.8. The development and function of TH1 cells. IL-12 is synthesized early in the response to intracellular pathogens by NK cells and dendritic cells, and drives the differentiation to TH1 cells, which synthesize IFN- γ and IL-2. T-bet is the lineage-determining transcription factor for TH1 development. MS = multiple sclerosis; RA = rheumatoid arthritis. ADCC = antibody-dependent cell-mediated cytotoxicity.

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TH2 cells

- Develop in the presence of IL-4 as a response to parasitic worms such as Helminthes and allergens, produced by mast cells, activated CD4⁺ T cells, and NKT cells → allergy
 - IL-4 activates Stat6 & GATA of T cells transcription factor
 - synthesize IL-4, IL-5, and IL-13
- IL-4, and IL-13 influence B cell class switch to IgE and IL-5 activates eosinophils, both characterize immune response to parasitic worms and type 1 hypersensitivity (allergy).

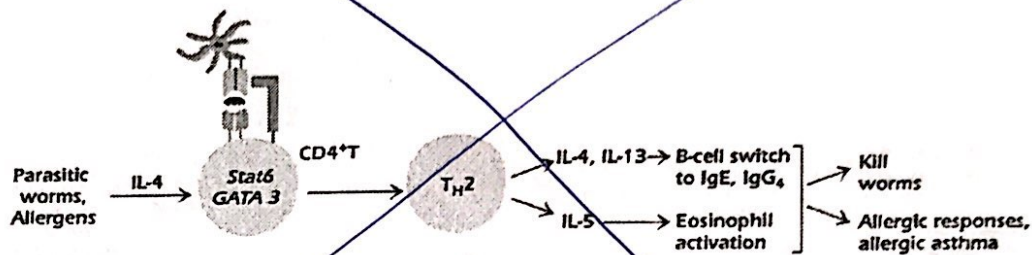


Figure 11.9. The development and function of T_H2 cells. The source of IL-4 that drives the differentiation to T_H2 cells is not resolved. Stat6 and GATA 3 are the lineage-determining transcription factors for T_H2 development.

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• T_H17 cells:

1-synthesize IL-17 and IL-22 cytokines.

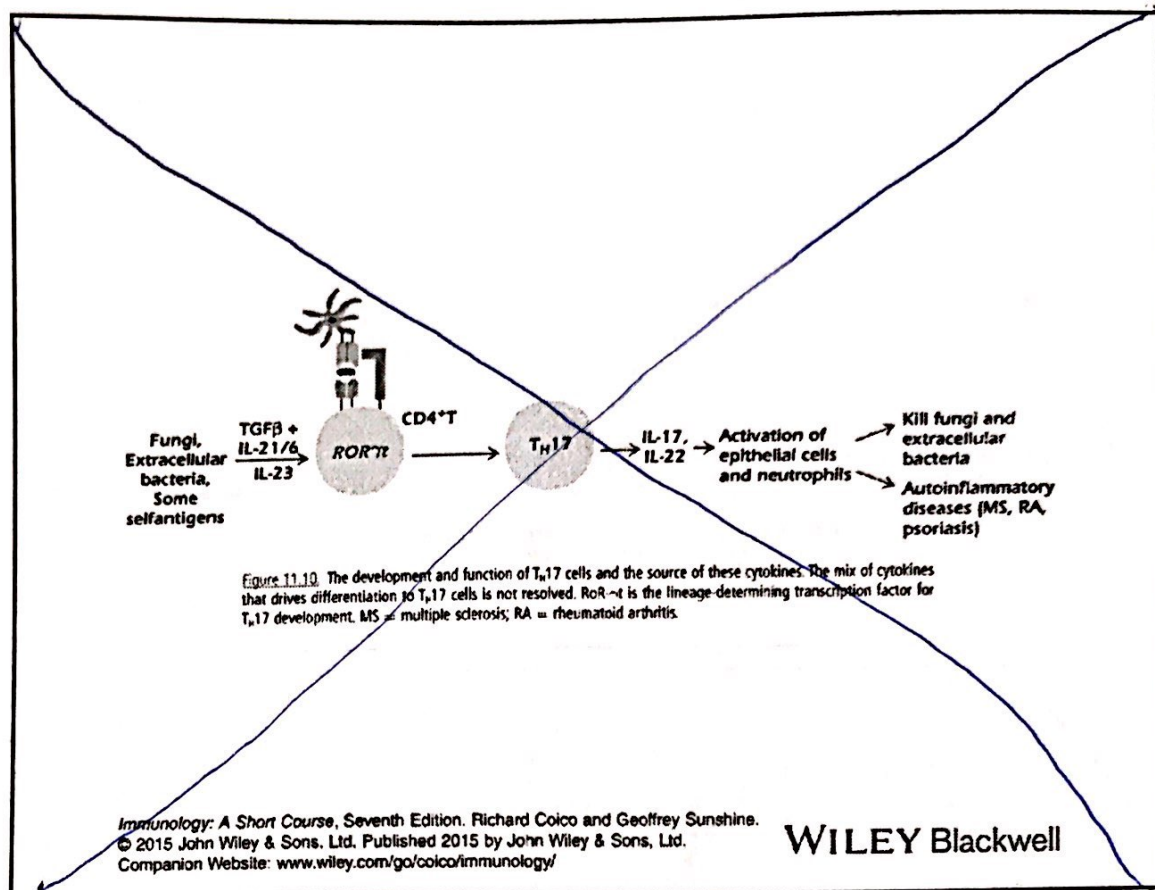
A. IL-17 and IL-22 are pro-inflammatory or they promote inflammatory response. *produce inflammation*

activate and recruit neutrophils and EC that secrete IL-1, IL-6 and TNF- α .

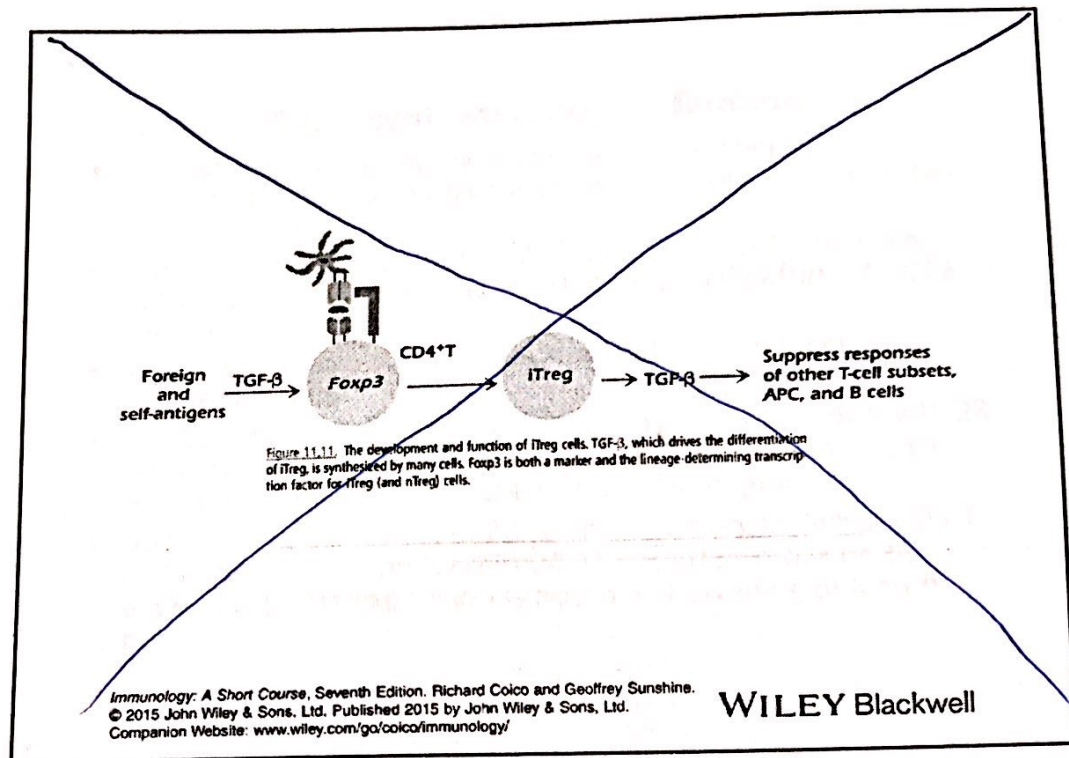
B. Th17 have been described in some autoimmune diseases (Rheumatoid arthritis, multiple sclerosis). *Th17 associate with*

2-Involved in responses to extracellular bac (*klebsiella pneumonia*) and fungi (*Candida albicans*). T_H1 and T_H2 have little effect on these pathogens.

3-Express high levels of receptor activator of NF-B ligand (RANKL) that activate osteoclasts which stimulate degradation of bone matrix



- characteristics*
- **T_{reg} cells:** subset of $CD4^+$ *دستگیر*
 - Express CD25 and CTLA4 *to isolate Treg cells we use antibody against CD25*
 - Suppress the differentiation of other $CD4^+$ subsets. They suppress also dendritic and B cells.
 - Suppress immune response directed against self molecules (Tolerance) and against foreign Ags (regulation). Lacking of T_{reg} cells (defective Foxp3 gene) results in autoimmune diseases like the x-linked IPEX syndrome and allergy.
 - This inhibitory action is due to secretion of inhibitory cytokines, IL-10 and TGF- β *(mechanism of inhibition)*



- Cross-inhibition of CD4+ cells

CK produced by one subset of CD4+ T cell inhibit the function of other subset.

1. IFN- γ synthesized by Th1 cells inhibits the development and function of Th2 and TH17 .
2. IL-4 synthesized by TH2 cells inhibit TH1 and TH17 cells
3. T_{reg} cells inhibit all the other TH1, TH2 and TH17 cells

4-IL-2 synthesis and proliferation

- Activated transcription factors enter the nucleus of the activated T cell and bind selectively to regulatory sequences of several genes.
- Two activated genes (within 24 hrs) are the IL-2 and IL-2R α (CD25) are transcribed and translated directly after B7-CD28 interaction.
- IL-2 is a T cell growth factor and act on the same cell or adjacent T cell
- The resting naive T cell expresses 2 chains β and γ of the IL-2R with low affinity. The synthesis of the IL-2R α chain due to activation converts the IL-2R into 3-chain high affinity R.
- Interaction of IL-2 with IL-2R results in enlargement of CD4 T cell and proliferation (within 24-48 hrs after initiation of activation). This expansion is noted as a swelling of lymph node

The activated T cell produce IL-2 and IL-2 receptor

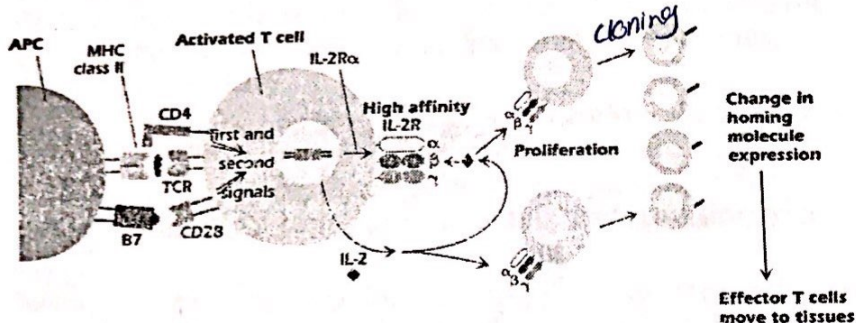


FIGURE 11.5 Activated CD4⁺ T cell synthesizes and secretes IL-2 and synthesizes the IL-2 receptor α chain. The interaction of IL-2 and the high-affinity IL-2 receptor results in proliferation of the CD4⁺ T-cell clone, change in expression of homing molecules, and movement of the resulting effector T cells out of the node and into tissues.

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* naive cells doesn't produce IL-2 nor IL-2R

* cloning happens when the cytokine and it's receptor are produced

* how can we differentiate between naive and activated TC?

we do elyza to determine IL-2 concentration in each

tube, if there's 0% concentration

that means that the tube has naive T cells

Clonal expansion and differentiation to effector cells

- As a result of secreted CK, enormous expansion of the original clone of Ag-specific cells up to 10,000 times the original clone size. This causes swelling of the L. node in which the response occurs.
 - The activated T cells differentiate into effector T cells. Activation of T cells results in a change in expression of adhesion molecules and chemokine R.
- A. Effector cells down-regulate the adhesion molecules L-selectin and CCR7 that allow the naïve T cells to enter the node.
- B. They up-regulate the adhesion molecules (**integrin**) and CCR10 that allow them to migrate out of the node to locations infected by pathogens.

5-B cell selection

- Generally, BCR recognize and bind free Ag in solution. Protein, polysaccharides or lipids may act as an Ag. The Ag that penetrates into tissues is carried to the lymph node draining the tissue.
- Thus, B cell epitopes are hydrophilic. Terminal side chain of Polysaccharides and hydrophilic portion of a protein molecules constitutes B cell epitopes (sequential) and conformational epitopes due to folded conformation of a protein.
- Among the trillions of different B cells, the Ag interacts in the L. node follicle (cortex) with a B cell expressing a specific BCR. The selected B cell moves to the T cell area (paracortex). The B cells that participate in response to an Ag, are called **follicular B cells**.

- The B cell captures Ag and presents it to the activated TH cell.

* B cell has the ability to present the Ag (could act as an APC)

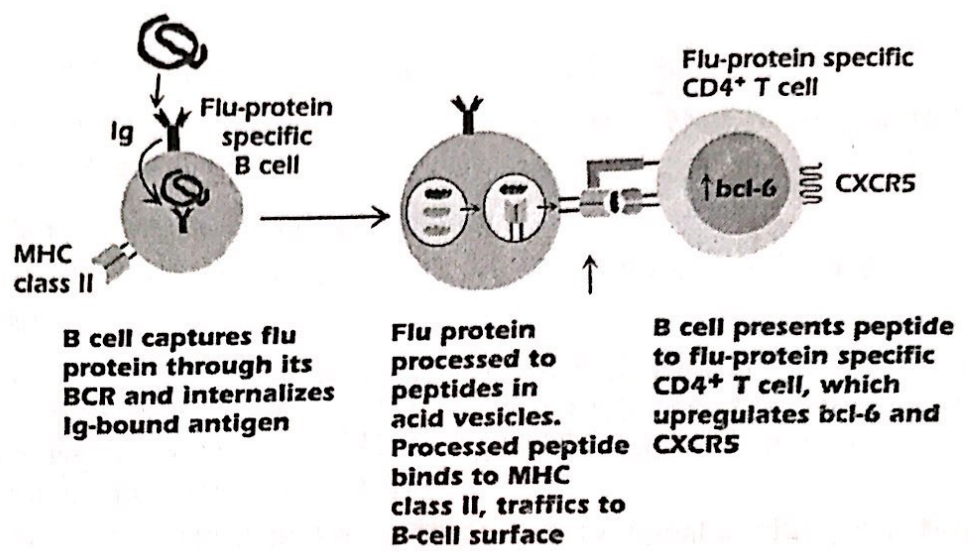


Figure 11.12. The B cell as an antigen-capture and antigen-presenting cell. A flu-specific B cell captures a flu protein antigen using its BCR, membrane Ig. The B cell processes the antigen and presents flu-protein peptides in association with MHC class II molecules to a flu-peptide specific CD4⁺ T cell. The interaction upregulates T cell expression of the transcriptional repressor, bcl-6, and CXCR5. These molecules characterize the T_{fh}.

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6_Cooperation between selected activated CD4⁺ and B cells

A. Development of T_{FH} follicular helper

Characteristics:

express the transcription factor Bcl-6, the chemokine receptor CXCR5 and secrete IL-21

CXCR5 is the marker for germinal center development and T-B cooperation

~~* T cell~~

* B cell doesn't need to be presented like T cell (T cell is presented by APC)

B. Paired interaction APC and CD4+ cell

- The interaction between B-T is similar to that between APC and CD4+ T cell.

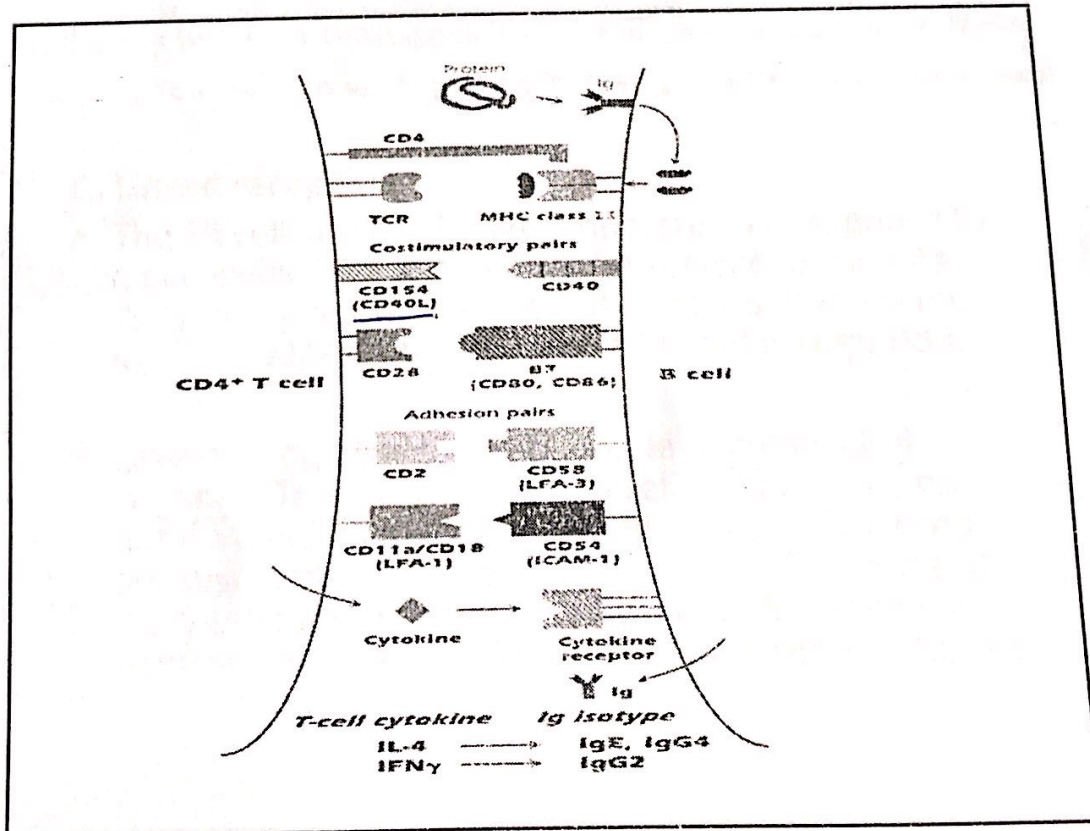
-Peptide/MHC on the B cell....TCR on T

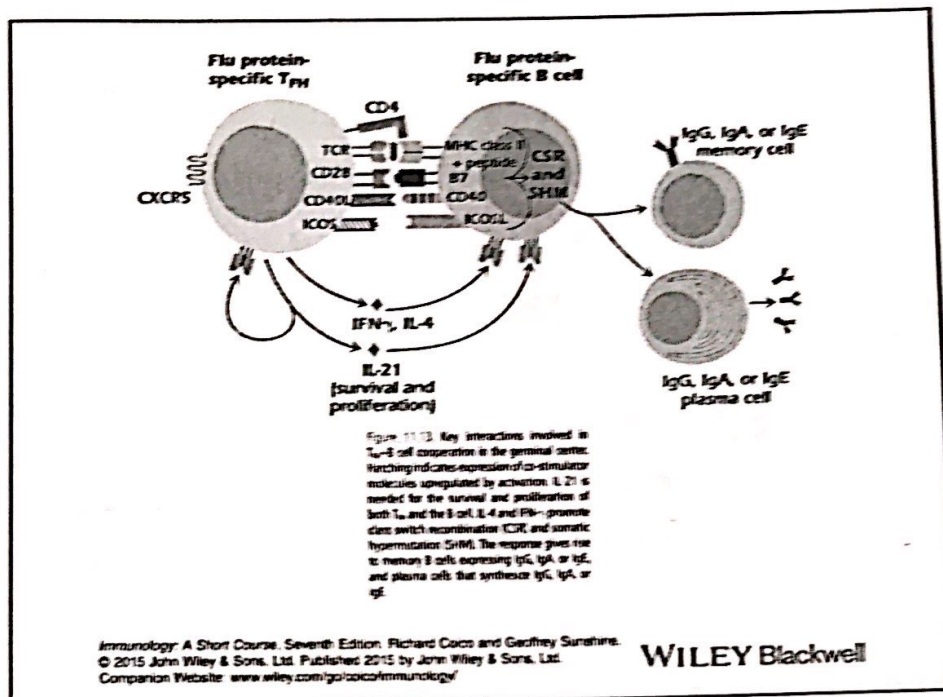
-Costimulatory molecules: B7—CD28, CD40—CD40L, ICOS—ICOSL

-Adhesion molecules

- This interaction upregulate T_{FH} cytokine synthesis which in turn affect the T cell itself and the B cell. IL-21 the signature cytokine of T_{FH} cells promote survival and proliferation of both T_{FH} and GC B cell and direct class switching e.g IL-4 promote IgG4
- The CD40-CD40L interaction appears critical in the activation of B cells. It is required for class switching. In hyper-IgM syndrome, boys with a nonfunctional CD40 or CD40L produce IgM Ab but no other isotype.
- People who lack a functional ICOS gene product do not develop normal germinal centers and are immunodeficient. (no GC development)

* germinal center: center for B cell ~~selection~~ development





* For cooperation to happen both epitopes must be covalently linked on the same Ag. To prove this a procedure was made:

C. Linked recognition

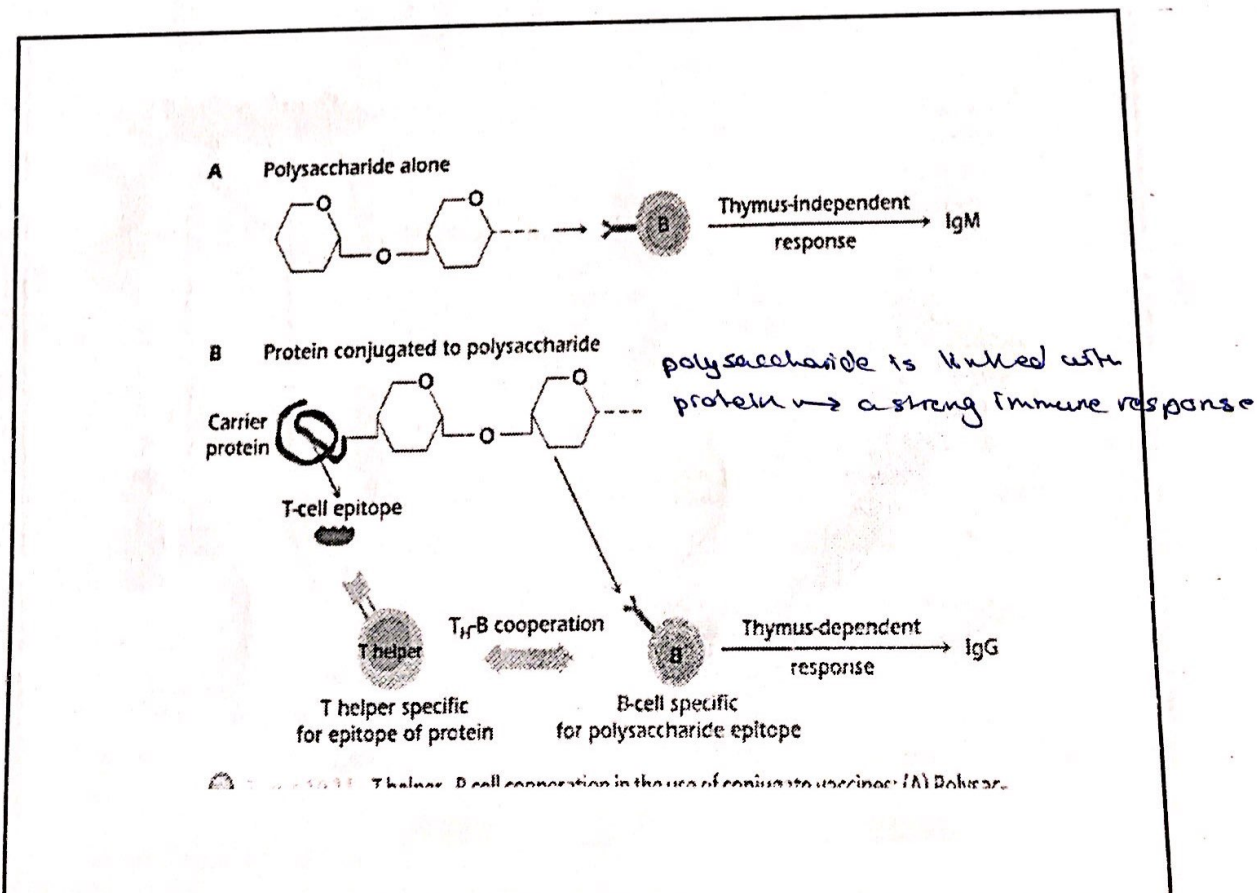
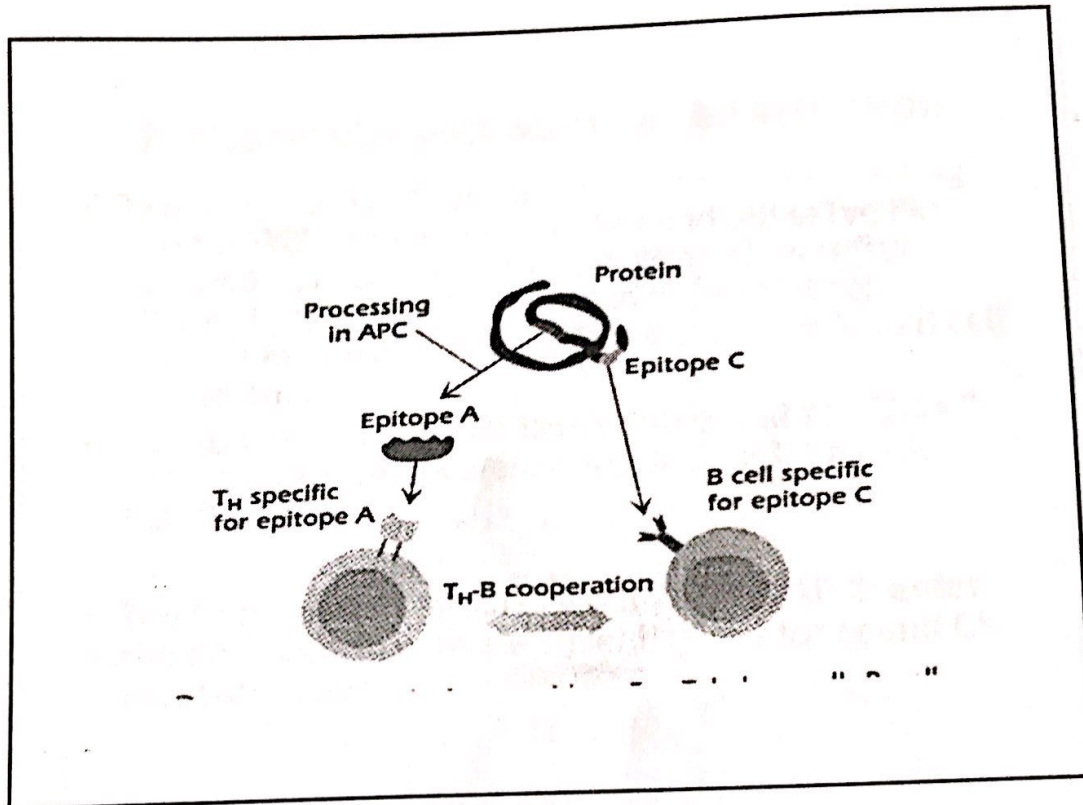
- The T_H cell and B cell that cooperate in a response to a particular TD Ag must both be specific for that Ag. They may respond to different epitopes of the same Ag. T-B cooperation is known as linked recognition.
- Linked recognition is the principle of conjugated vaccines. The marginal zone B cells do not develop until a child is about 2 years old. So young children are susceptible to infections from *H. influenzae* and *S. pneumoniae*. Purified polysaccharide from the bac is physically linked to a carrier protein as tetanus toxoid

* procedure:

epitope A and epitope B were isolated

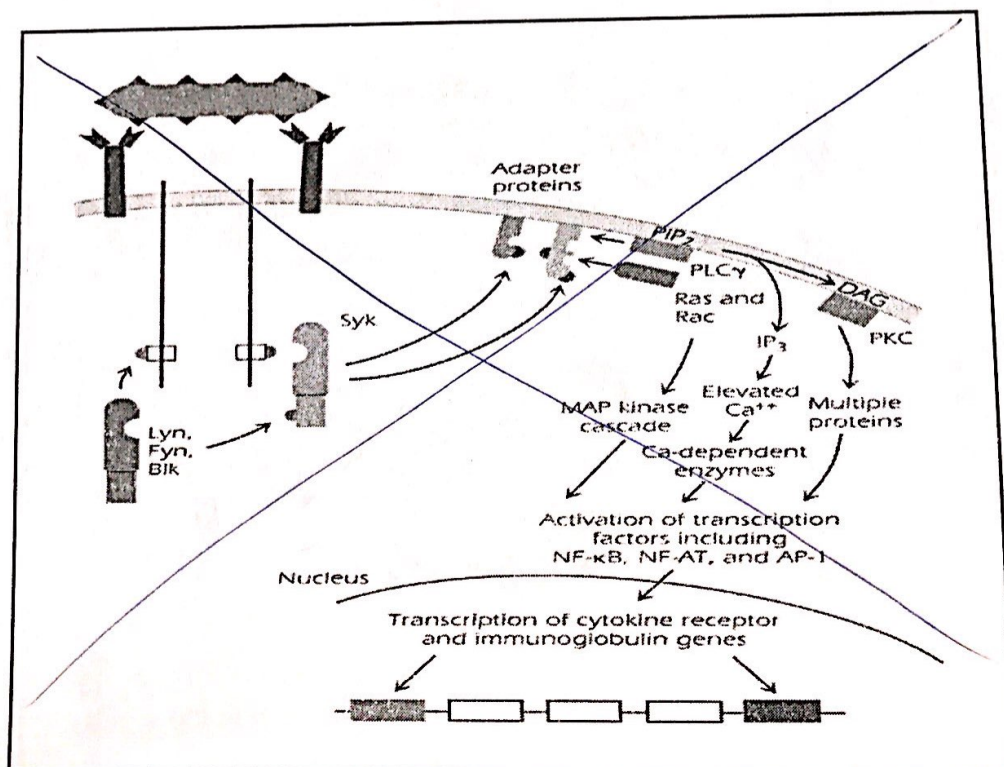
epitope A and epitope B were injected in the same mouse

no activation happened (because they're not linked)



7-Intracellular pathway in B cell activation

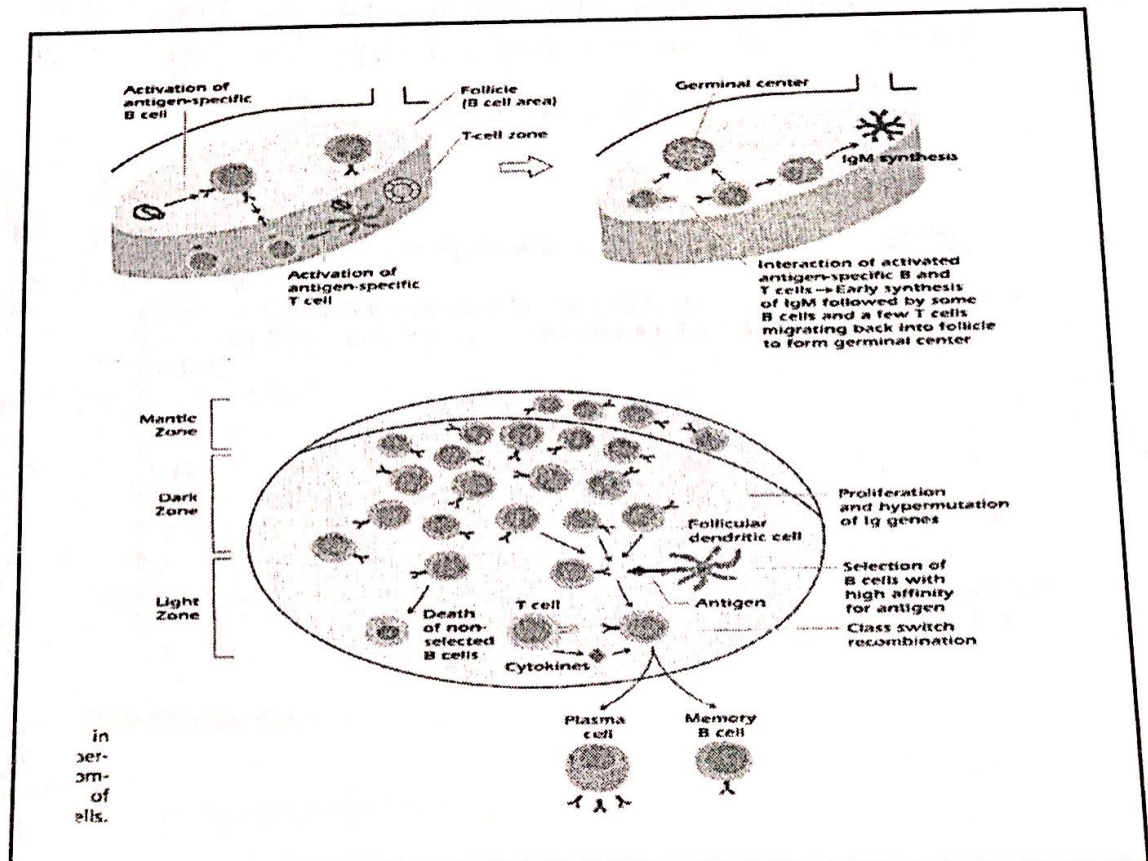
- The first signal includes interaction between the Ag and the BCR. The associated Ig α and Ig β serve like CD3 and CD4 as t.d. molecules that activate Tyr kinases Lyn, Fyn and Blk. This in turn activates another kinase Syk which plays a central role in B cell activation.
- The subsequent reactions resemble that for CD4+ T cells. Adaptor molecules, PL- γ , DAG, IP₃, Ras, rac, MAP kinase.
- The transcription factors NF- κ B, NF-AT, AP-1 enter the nucleus and activate several genes for Ig and CK receptor genes.



7-Events in the germinal centers of activated B cell

- In the germinal center, the B cell undergoes:
 - somatic hypermutation in which B cells with Ig variable region genes that synthesize Ab with higher affinity for the activating Ags are selected.
 - Class switching
 - Plasma cells and memory cells are developed from B cells (irreversible)
- The activated B cell proliferate rapidly in the dark zone of germinal center.
 - B cells that do not proliferate are pushed out of the area and form a mantle zone. These undergo somatic hypermutation.
 - B cells that receive no survival signals from Th or follicular dendritic cells die in the light zone.
 - The CK influence the class switching process.
 - The activated B cells by Th cells may differentiate to become either a plasma cell or memory cell.

* plasma cell
lose membrane
proteins



T-Independent B cell response

- Most Ags are known as thymus-de. (TD) Ags because they require Th cells. The thymus-indep Ags (TI) the responding B cells do not require Th cells.
- Features of i.r.
 - a. i.r occurs in people or animals that lack a functional thymus or T cells (immunodeficiencies) → no class switch, no memory → no T cells
 - b. generate IgM and do not give rise to memory cells.
 - c. cross-linking of BCR with the repetitive epitopes directly activates the B cells
- TI Ags are divided into 2 classes:
 - TI-1 Ags: repeating structure (repetitive epitopes)
 - mitogenic at high conc.
 - polyclonal activator e.g LPS of G- bacteria like E. coli
 - The protein coat of some viruses as polio virus
 - TI-2 Ags: are not mitogenic at high conc.
 - like bac and fungal polysaccharide as dextran, ficoll capsule polys. Of *Hemophilus influenzae* and *Strep pneumonia*

*majority of cells are dependent

*polysaccharide activate only B cells because it's a repetitive epitope

*procedure to ~~is~~ TD and TI:
two tubes, TDAg, TIAg

we inject two mice each with one sample (mouse 1 → TDAg, mouse 2 → TIAg)
then we inject them again a week later, the mouse that doesn't develop an immune response is the mouse injected with TDAg (no memory)
to further determination we take blood sample of each mice and inject it with Ab against IgG, the one without IgG is TDAg -

9-Other ways to activate B or T cells

1. SuperAgs:

Bac toxins as *S aureaus*.

Activate CD4+ cells expressing a particular V molecule as one chain of TCR, regardless of chain.

a high % of T cells of various specificities may become activated in the presence of superAg. The massive release of CK following SuperAg can result in injury to the host.

2. Plant proteins:

Polyclonal activators for B cells include LPS

Polyclonal activators for T cells include: the plant glycoproteins

Phytohemagglutinin (PHA) and Concanavalin A (ConA) which are lectins.

Polyclonal activators for B and T cells include: pokeweed mitogen (PWM).

Because the B or T-cell response of healthy human blood to LPS or ConA (PHA) falls in a well-defined range, a low response frequently indicates that a person is immunosuppressed.

3. Antibodies: Abs against CD3 activate T cells to proliferate.

procedure to detect immune deficiency!

blood sample put in

two flasks

1 → conA (positive control)

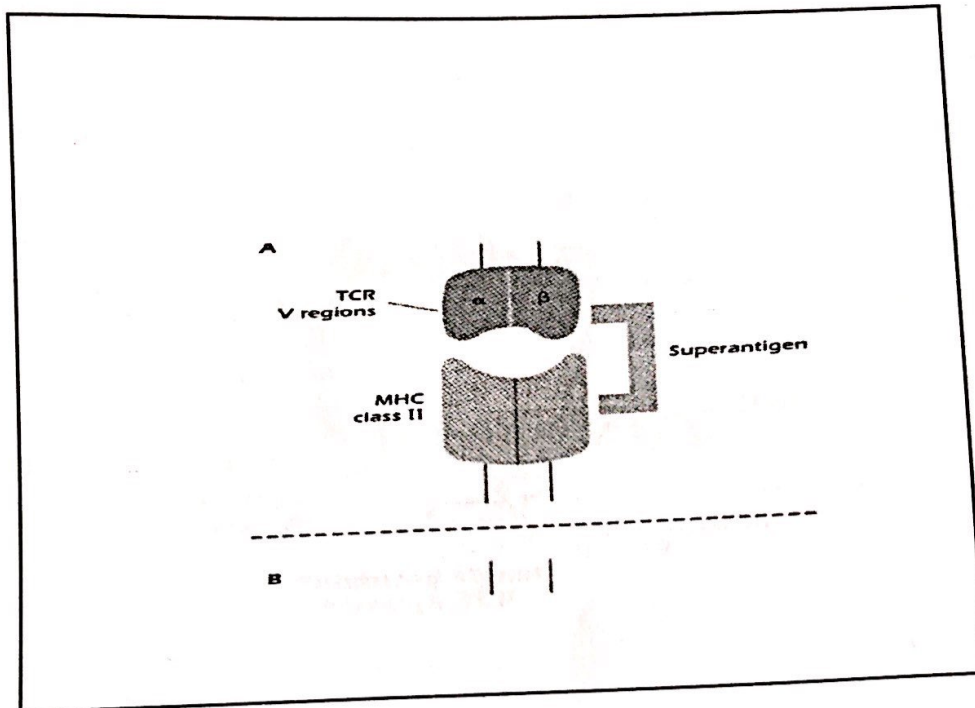
no activation of TC?

immune deficiency

2 → LPS + polysaccharide

no activation of BC?

immune deficiency



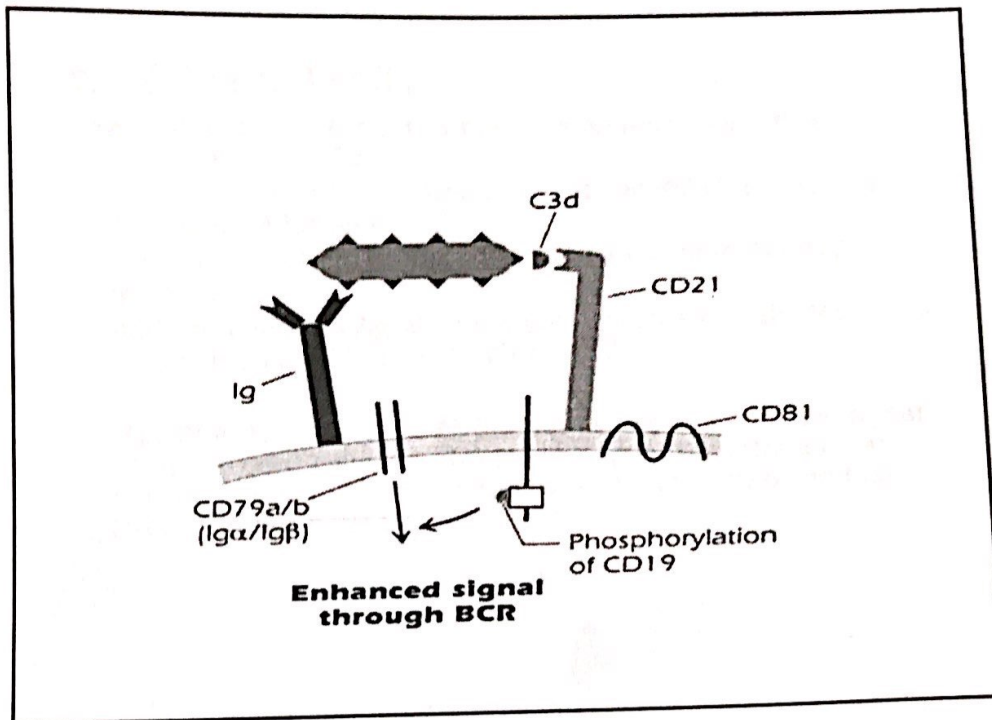
10-Modulation of BCR Signal (+, -)

- **Positive modulation**

Like coreceptors of TCR (CD4 or CD8), the BCR coreceptors (CD81, CD21, CD19) decreases the amount of Ag needed to trigger the B cell (decrease the threshold) 100-1000 fold

- **Microbial pathogens activate the complement system and become coated with complement component C3dg.**

The R for C3dg is CD21. Thus the pathogen bound to C3dg becomes attached via CD21 to the B cell. The cross-linking of BCR-Ag-C3dg-CD21 deliver an activating signal to B cell through phosphorylation of CD19.



• 11-Negative modulation

a. CD28----CTLA4

CTLA4 competes with CD28 for B7 binding and has a higher affinity than CD28, the cytoplasmic domain has ITIM motif, and turns off the production of IL-12

b. Programmed cell death protein PD-1

- expressed on activated T cells.
- Inhibit ZAP-70 function
- Binds to PDL (expressed on most tumor cells and many normal cells)
- Transmit negative signals into T cells.
- used in clinical trials to treat some autoimmune diseases

c. CD32 on B cells

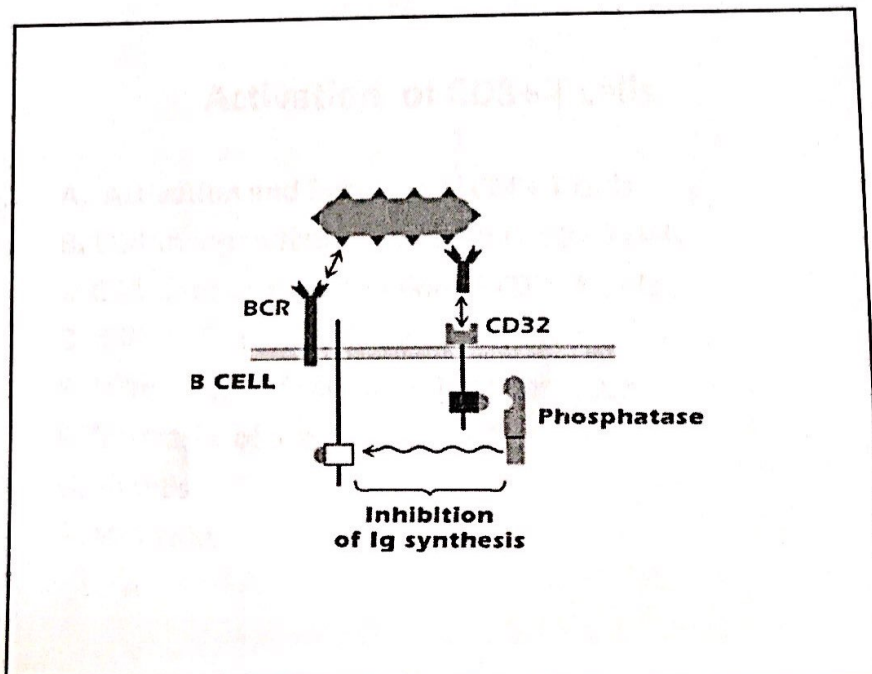
The produced Ab after overcoming the Ag inhibits a further b cell response to that Ag.

-can be induced by injection of Ab into an individual shortly before or during an i.r.

-This property has been used to treat Rh incompatibility reaction

-IgG aggregates or Ag-Ab complexes interact with the low affinity FcγR (CD32) expressed on B cells.

-The BCR-Ag-Ab complexes-CD32 give rise to a negative signal. This activates a phosphatase to the Tyr of the intracellular portion of CD32 (ITIM) that removes the phosphate leading to an inhibitory signal



Activation and Function of CD8+ T cells: Cell-mediated Immunity

Activation of CD8+ T cells

- A. Activation and Function of CD8+ T cells**
- B. CD4-Independent activation of CD8+ T cells**
- C. CD4-dependent activation of CD8+ T cells**
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Activation and Function of CD8+ T cells: Cell-mediated Immunity

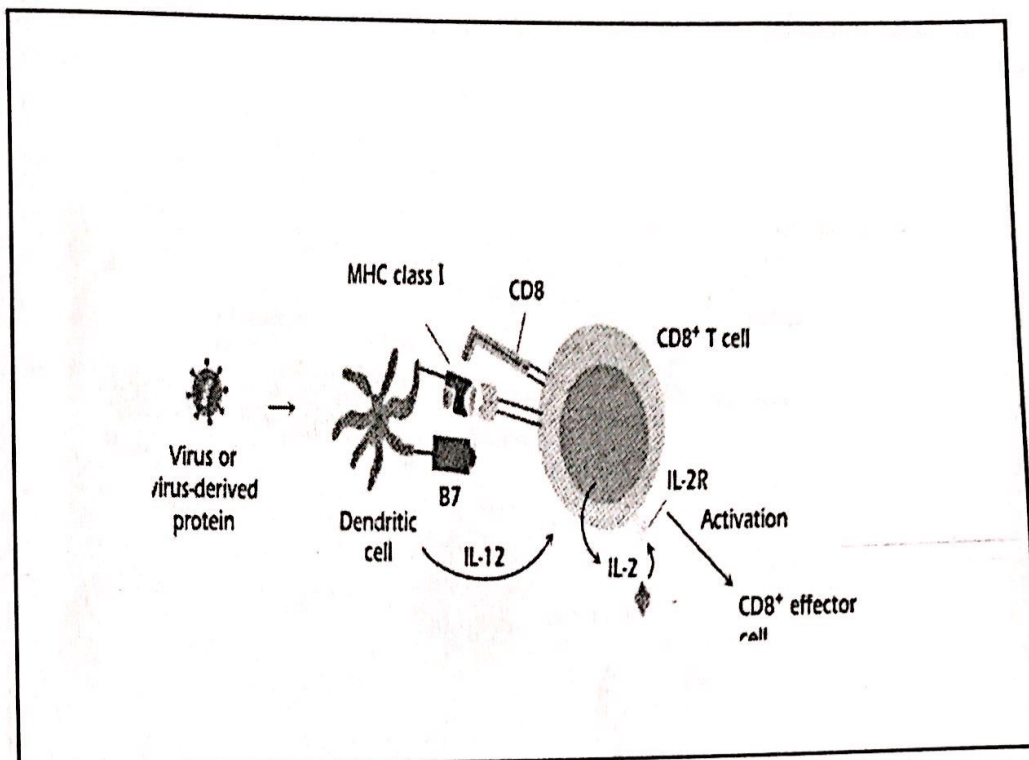
- The principle function of CD8+ T cell is to kill host cells that have been infected by viruses and bac.
- Also involved in killing transplanted foreign cells and tumor cells.
- Therefore, CD8+ T cell is referred to as Cytotoxic T lymphocyte (CTL).
- The target cell can be any cell in the body.

Generation of Effector CD8+ T cells

- Two pathways are known for the activation of CD8+ T cells by viruses.

CD4-Independent activation

- A dendritic cell can present a virus-derived peptide associated with an MHC class I molecule.
- Dendritic cell provides the first (peptide/MHC...TCR) and second signal (costimulatory molecules) as well as secretion of IL-12 to activate CD8+ T cell.
- The CD8+ T cell synthesize IL-2 that induces proliferation of the activated T cells.



CD4+-dependent activation

This activation into effector cells follow many of the same principles described for the generation of CD4⁺ T cells.

Activation of CD8⁺ T cell

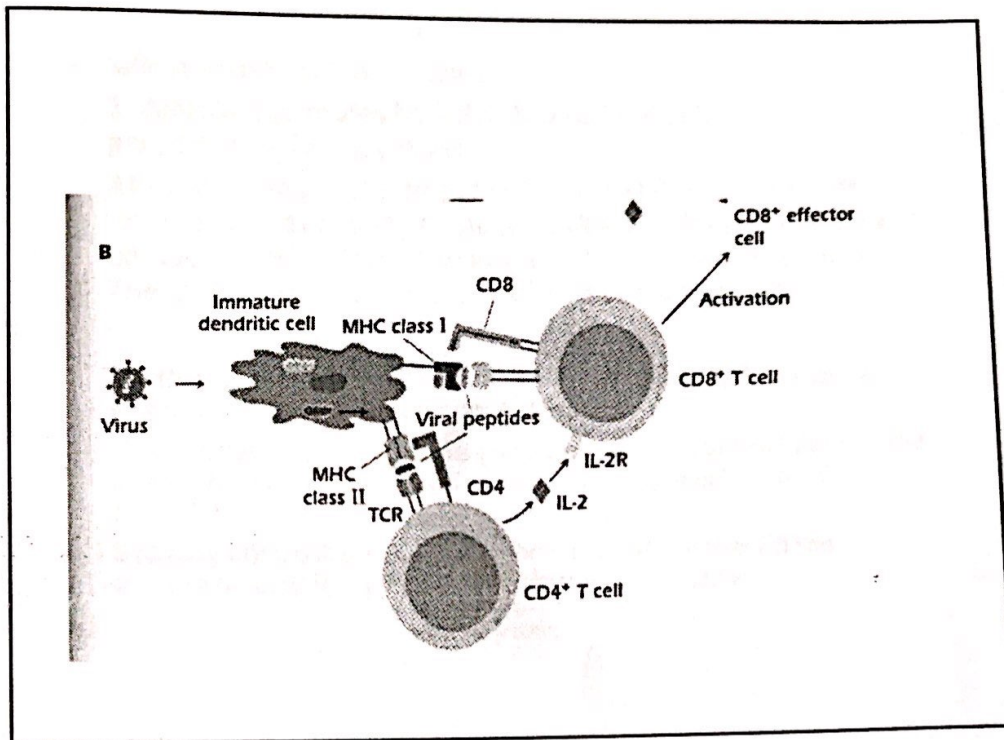
-A dendritic cell that has taken the virus as an exogenous Ag presents virus-derived peptides in association with MHC class II molecules to a virus-specific CD4⁺ T cell. Activated CD4⁺ T cell proliferate and synthesizes IL-2.

-The same or another dendritic cell may also present viral Ags to CD8⁺ T cells in association with MHC class I molecules.

The virus epitope that activates CD4⁺ T cell is different from the epitope that activates the CD8⁺ T cell.

-The third signal involve secretion of CK that induce proliferation and activation of the selected CD8⁺ T cell

-Intracellular activation of CD8⁺ T cell is the same as CD4⁺ T cell include the Tyr Kinases (Fyn, Lck, ZAP-70, and others).



CD8⁺ T cell –Killing of target cells

Cell to cell contact: Once activated, the CD8⁺ T cell initiates killing by attaching to the target cell.

The cell to cell contact is critical for the killing process. In agar immunodiffusion experiment, if the activated CD8⁺ T cells are put in a well and the other well is filled with the target cells, no killing will be achieved.

The paired adhesion molecules expressed on the T cell and target cell surfaces help to maintain contact for several hours.

- Killing occurs by 2 pathways:

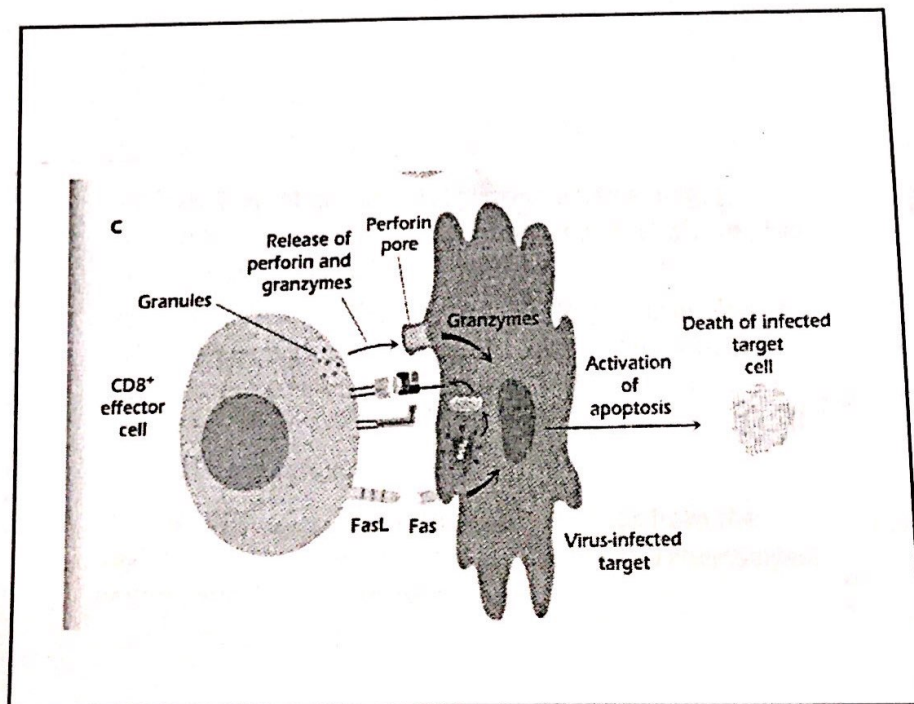
1. **cytotoxic granules by activated CD8+ T cell:** the predominate killing pathway.

After attaching to the target cell, the T cell reorganizes its internal structure so that the granules are close to the area of contact. T cells release the granule contents by exocytosis. The major constituents are perforins and granzymes.

Perforins: are transmembrane channels form pores in the target cell membrane similar to complement system.

Granzymes: are ser proteases that pass into the target cell through the pores. Granzymes interact with intracellular components and induce apoptosis.

Granulysin: small protein that enter the target cell that kill the intracellular pathogens such as *Mycobacteria* and *Listeria*



2. Fas-FasL interaction: A second killing pathway

Fas: surface molecule expressed on many host cells

FasL: expressed on activated CD8+ T cell

This interaction activates apoptosis of the target cell via a sequential activation of caspases (a proteolytic enzymes inside the target cell). The cell dies within hours.

Killing by apoptosis prevent the spread of the virus into other cells.

The killing is logarithmic as the T cell detaches from the target cell to attack and kill additional target cells.

Activation of CD8+ T cell and killing of the target cell are separate events

- cytotoxic T lymphocytes (CTL) possess a fast-acting, perforin/granzyme-mediated, as well as a much slower, Fas ligand (FasL) -driven killing mechanism
- memory CD8+ T cells that release the DC-activating factor TNF- α before the release of cytotoxic granules induce DC expression of an endogenous granzyme B inhibitor PI-9 and protect DCs from CTL killing with similar efficacy as CD4+ Th cells.
- Memory CD8+, CD4+, Dendritic cells produce from the beginning protease inhibitor to protect DC and themselves, before secretion of cytotoxic granules.

MHC restriction and CD8+ T cell killer function

- **MHC restriction:** CTL recognizes the combination of Ag (peptide) plus MHC molecules rather than Ag alone.
A virus-specific CD8+ CTL recognizes and subsequently kills a target cell expressing a specific combination of viral peptide and a particular MHC class I molecule.
- A CD8+ CTL specific for a flu virus peptide and HLA-A2 kills a target cell expressing HLA-A2 that has bound the flu-derived peptide. In the absence of the flu peptide, this CTL does not kill uninfected or normal cells from an individual expressing HLA-A2. In addition, this virus-specific CTL does not kill targets expressing different combinations of peptides plus MHC molecules such as measles-virus derived peptide bound to HLA-A2 or even the flu peptide bound to different HLA molecule.

Killing of any infected cell:

- The recognition occurs regardless of the expression of any MHC class II molecule by the target cell. MHC class I are expressed on almost every cell in the body. Since the infection pathogen may infect any nucleated cell, a pathogen generates peptides that associate with MHC class I molecules expressed on the surface of any nucleated cell. Thus, killing by CTL provides a mechanism to eliminate any cell in the body that becomes infected with a pathogen.
- Elimination of the pathogen results in the destruction of host cells, but this is the price the individual pays for removal of the source of the infection.
- Harmless or noninfectious Ags do not generally trigger CTL response. In contrast, intracellular pathogens evoke CTL (CD8+) and Th (CD4+) cells and induce Ab synthesis

Memory T cells

1. Ability to survive in a quiescent state after Ag is eliminated. Naïve cells live for weeks but memory cells survive for months or years
2. The No of specific memory T cells is greater than the No of specific naïve cells before Ag encounter.
3. Express increased levels of antiapoptotic proteins (Bcl2) for prolonged survival
4. Undergo slow proliferation depending on cytokines (IL-7) but not Ag recognition
5. Receptors equivocal but have R for IL-7 and CD27
6. Depending on homing properties:
 - a. Central memory: Express CCR7 and L-selectin and home mainly to lymph nodes
 - b. Effector memory: do not express CCR7 and selectin and home to mucosal tissues (peripheral)

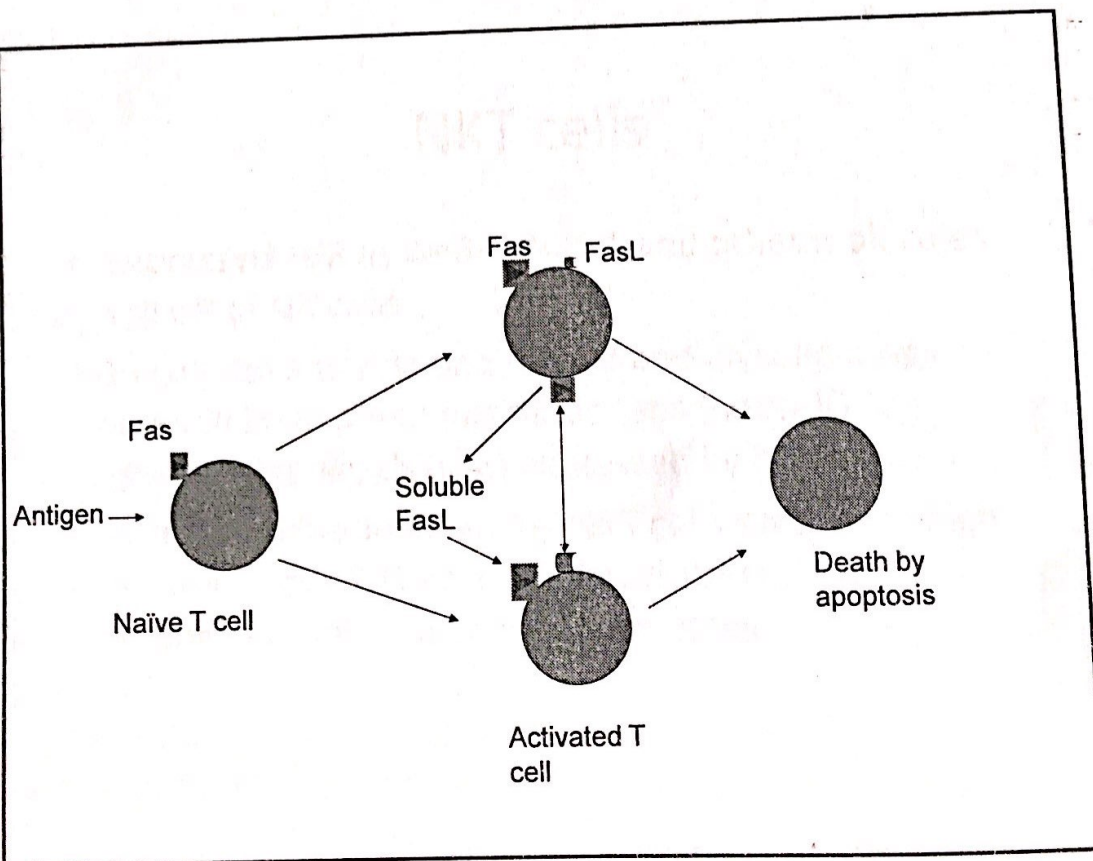
Traffic of activated CTL

- Unlike naïve CTL which circulate only through lymph nodes and other secondary L. organs, effector and memory CTL can enter tissues.
- Their ability to enter tissues is governed by the expression of adhesion and chemokine receptors. They downregulate the expression of CCR7 and L-selectin expressed by naïve CTL and upregulate others.

Termination of the CTL response

After repetitive stimulation, CTL cells become susceptible to apoptosis and the majority of activate CTL cells are eliminated by **ACTIVATION-INDUCED CELL DEATH (AICD)** mediated by **Fas-FasL**.

- a. Activated CTL express both Fas and FasL and they kill each other once Ag has been removed.
- b. Soluble FasL protein is generated by Activated CTL which binds to the Fas leading to apoptosis



Activated Macrophages

- Two distinct subsets of macrophages have been proposed, including classically activated (M1) and alternatively activated (M2) macrophages.
- CD4⁺ Th1 cells activate macrophages by contact-mediated signals delivered by CD40L-CD40 interactions and by CK.
- Immunity against *Listeria monocytogenes*. The bac were killed not by T cells from immune animals but by activated Mac.
- Activated macrophages kill phagocytosed microbes mainly by the actions of reactive oxygen species, NO and lysosomal enzymes
- **M1 macrophages are activated by LPS, IFN- γ , or TNF- α are always activated by IL-4 and IL-13.**
- M1 macrophages express a series of proinflammatory cytokines, chemokines, and effector molecules, such as IL-12, IL-23, TNF- α , iNOS and MHCI/II.
- M2 macrophages express a wide array of anti-inflammatory molecules, such as IL-10, TGF- β , and arginase1. In most tumors, the infiltrated macrophages are considered to be of the M2 phenotype, which provides an immunosuppressive microenvironment for tumor growth.

NKT cells

- Expressed TCR as well as NK1.1 and other molecules typical of NK cells.
- Recognize and respond to lipid and glycolipid Ags derived from infectious pathogens (nonself) and glycosphingolipids (self) expressed by host cells.
- After stimulation by an Ag, NKT cells synthesize high levels of both Th1 and Th2 associated CK. So it is important in the early clearance of bac

$\gamma\delta$ cells

- Found predominately at mucosal epithelial sites to provide the first line of defense against invading pathogens
- Produced CK e.g IFN- γ and have cytotoxic function.
- Do not respond usually to peptide associated with MHC molecules but do respond to phospholipids and other phosphoantigens